

An Ensemble Deep Learning Approach for Leukemia Stage Classification from Microscopic Blood Cell Images

ABSTRACT

Leukemia remains a life-threatening blood cancer worldwide, where accurate staging at an early stage plays a crucial role in the success of patient treatment. Conventional microscopic blood cell observation is highly expert-dependent, time-consuming, and subject to interpretation variability. To overcome these limitations, in this study, we propose an ensemble deep learning framework to automatically identify four stages **Benign**, **Early Pre-B**, **Pre-B**, and **Pro-B** from microscopic Peripheral Blood Smear (PBS) images. First, preprocessing, normalization, augmentation, and balancing techniques are applied to address the class-imbalance and image-variation issues present in the dataset. Then, two different feature-detecting CNN architectures, ResNet18 and VGG16, are combined to create an ensemble model that is capable of learning both high-level semantic features and fine-grained local features. Compared with widely used baseline models such as **EfficientNet-B0**, **MobileNetV3** and **DenseNet121**, the proposed **Ensembled ResNetVGG** model achieved the highest test accuracy of 99.49%. To clearly illustrate how the model makes decisions, a saliency map was generated using the Guided Backpropagation method. This map clearly shows the most important cellular components in determining the stage of leukemia. According to the research results, the proposed ensemble method not only provides a high level of accuracy, but also serves as an interpretable and reliable diagnostic aid for use in a real clinical setting.

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Chapter 1

Introduction

This chapter briefly outlines the necessary conditions, research goals, and key points for how to quickly and accurately identify the four stages of leukemia from PBS images using the proposed ensemble deep learning method.

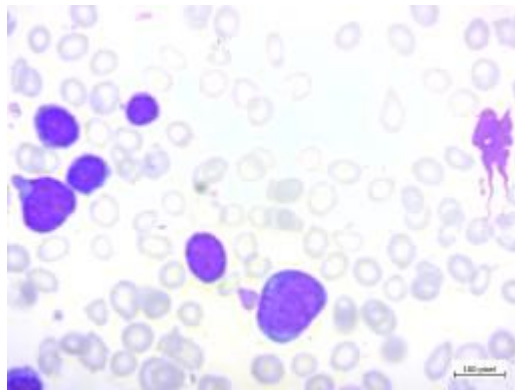
1.1 Introduction

Leukemia is one of the fastest-spreading and most deadly blood cancers in the world. It is very important to accurately diagnose the stage of the disease at the beginning, because it determines the patient's treatment plan and long-term survival. In the early stages of the disease, very subtle and vague changes are seen in the cells, which are not easily understood. Therefore, in most cases, the diagnosis is delayed. This makes treatment more complicated and if measures are not taken at the right time, the success of the treatment is greatly reduced. In the Peripheral Blood Smear (PBS) image under the microscope, the cell size, structure of the nucleus, characteristics of the cytoplasm and small changes in the cell size play a very important role in identifying the four stages of leukemia (Benign, Early Pre-B, Pre-B and Pro-B).

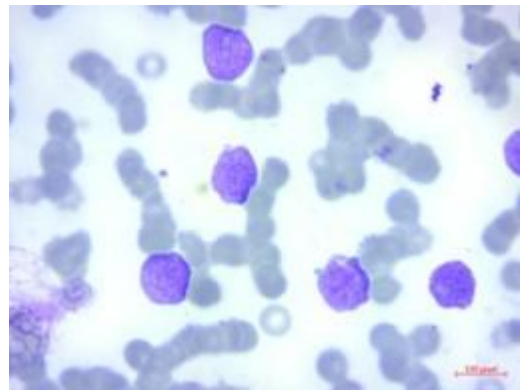
Conventional diagnosis relies on fully trained hematopathologists. This analysis takes a long time, can be interpreted differently by different experts, and is prone to human error. Cells at different stages of leukemia look very similar—especially in the Pre-B and Pro-B stages—which makes it easy to misjudge the stage. Therefore, automated, reliable, and easy-to-understand technology-based decision-support systems are now essential in modern medicine. In today's healthcare, there is a lot of clinical data, microscopic images of blood cells, lab reports, and digital records—all of which are important ingredients for building AI-based models. But in reality, this data is often uneven, with more or less information in some categories, different labs have different image quality, and the number of samples may be small. For these reasons, a single model cannot work properly. At the same time, it is very important for doctors to understand—which features the model is looking at to determine which stage.

Recently, deep learning—especially Convolutional Neural Network (CNN)—has shown great success in microscopic cell analysis. But a single model often fails to capture the subtle differences between cells. Therefore, ensemble methods, which use multiple CNN models together, can extract more stable, reliable, and diverse features. This method is particularly effective in identifying the stage of leukemia. In this study, we developed an ensemble deep learning method that combines two different convolutional neural networks—ResNet18 and VGG16—to form a robust and reliable model. Data preprocessing, augmentation, normalization, and data balancing were performed to maintain the diversity and class balance of the images, which increased the quality of the

data. After training, this ensemble model was able to identify the four stages of leukemia very accurately. This can play an important role in diagnosis, treatment planning, and risk assessment. To understand how the model is making decisions, we created a saliency map using guided backpropagation. This map shows which cellular features are most important to the model. This makes the use of AI in the medical field more credible and interpretable, so that doctors can easily understand the model's decisions.



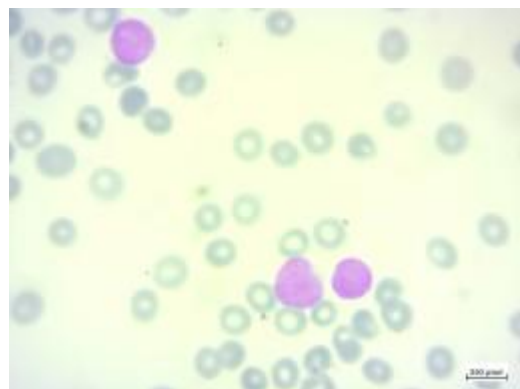
Sample Image of Benign Stage



Sample Image of Early Pre_B Stage



Sample Image of Pre-B Stage



Sample Image of Pro-B Stage

Fig-1.1: Sample Images of the Microscopic Dataset

To understand how the model is making decisions, we created a saliency map using guided backpropagation. This map shows which cellular features are most important to the model. This makes the use of AI in the medical field more credible and interpretable, so that doctors can easily understand the model's decisions.

Key contributions:

1. We have developed a powerful ensemble CNN model using ResNet18 and VGG16 to detect leukemia stage.
2. Preprocessing, data augmentation, and balancing have been scientifically performed to enhance the quality of the data.
3. It can perform very accurately on a variety of images and is able to detect even small cellular differences.

4. The model is interpretable using guided backpropagation, meaning doctors can easily understand why the model made the decision it did.
5. The performance of this model is much better than the baseline model, which is more reliable for future clinical use.

1.2 Motivation

1. Clinical Reliability and Early Diagnosis

The main goal of this research is to make accurate diagnosis of leukemia more reliable, easier and faster. Determining the stage of the disease by looking at small changes in blood cells is extremely difficult even for experienced doctors. If the stage is wrong, it can lead to treatment delays, decision errors and even loss of life of the patient. Therefore, a technology that can quickly, automatically and with high accuracy identify the stage of leukemia is highly needed. The proposed ensemble deep learning model is designed to meet this need.

2. Challenges of Manual Microscopic Diagnosis

Slide analysis of blood cells is usually completely dependent on human experience. Small changes in cell size, nuclear structure and cytoplasm are difficult to detect with the naked eye and are prone to errors. In particular, cells in the pre-B, pre-B II and pro-B stages are very similar, so manual staging is more likely to be wrong. These problems are the main motivation for our research to create a more scientific, reliable, and automated AI-based solution.

3. Data Imbalance and Real-World Variability

Our goal is to create a robust, balanced, and accurate model for identifying the different stages of leukemia, which will also work well on real-life images. The problems are that some stages have many images, some have few, and different labs or microscopes can show images of the same cells differently.

4. Identify the problem

Some stages in the dataset have more images, some less the model may be biased. Different labs, microscopes, and staining techniques can cause images of the same cell to look different. Goal: Create a robust, balanced model that will perform well in real-world settings.

5. Using AI in medical imaging

A single CNN cannot always capture subtle differences in cells. The high-level features of ResNet18 and the fine texture capabilities of VGG16 are combined. As a result, doctors will receive accurate and effective help.

6. Explainability

Using guided backpropagation (GBP), the model can be seen to which cellular parts the model is paying more attention. This gives doctors confidence and makes the model credible for clinical use.

7. Clinical benefits

Hematology labs have to examine many slides for more time, higher risk of error. Automated AI models will reduce workload, reduce errors, and provide patient results faster.

8. Future implementation and scalability

If the model is stable and accurate can be easily used in hospitals, labs, or clinical systems. In the long run, the treatment process will be faster, more accurate, and more standardized.

1.3 Objectives

1. Building an Advanced Ensemble Deep Learning Model

An ensemble model was created by combining two powerful CNN models, ResNet18 and VGG16. It combines high-level features and fine texture detection capabilities. As a result, leukemia stages can be identified more accurately, unbiased, and stably.

2. Pre-processing and data balancing

Various pre-processing steps are used to increase image diversity, class balance, and reduce noise, such as: Normalization, Data augmentation, Data balancing. This helps the model better understand and learn from real-world PBS images.

3. Fine cellular feature detection

Pre-B and pro-B cells look very similar, so it is difficult to distinguish them. Advanced CNN feature extraction allows for better detection of the shape, structure, and other fine differences in the nucleus.

4. Enhancing the interpretability of the model and establishing clinical confidence

When using AI in medicine, high accuracy alone is not enough; doctors need to know what information the model is based on to make decisions. Therefore, saliency maps were created using directed backpropagation, which shows which cellular regions are most influential in the decision-making process. This increases the transparency of the model and increases the confidence of doctors in its clinical use.

5. Comparative analysis with multiple modern deep learning models

To evaluate the performance of the ensemble model, it was compared with modern CNN architectures such as MobileNetV3, EfficientNet-B0 and DenseNet121. This

analysis shows where the model excels, where there is room for improvement and how suitable it is for use in real medical practice.

6. Evaluation of the accuracy, stability and generalization ability of the model

The performance of the model was measured using various evaluation indicators including accuracy, precision, recall, F1-score and confusion matrix. This shows how quickly the model learns, how stably it works on different datasets and how much it can generalize to unknown real-world situations.

7. Proposing a Clinically Usable Solution for Automated Leukemia Staging

The ultimate goal of the research is to create a robust, reliable, and easy-to-use AI-based system that can be easily integrated into hospital or laboratory decision-support systems in the future. This will make doctors' work easier, reduce human error, and provide patients with fast and accurate results.

1.4 Methodology

This study developed a powerful and accurate AI model that can automatically identify four stages of leukemia—benign, early pre-B, pre-B, and pro-B from peripheral blood smear images taken under a microscope. The data was carefully selected, cleaned, and pre-processed before use. Image transformation, normalization, rotation, flip, and color correction were performed to address issues such as images from different labs, color differences, and uneven number of classes. In addition, the dataset was balanced by increasing the number of images in classes with fewer images. The data was split in a ratio of 80/20 and 70/30, so that training was stable and reliable results were obtained.

Two powerful CNN ResNet18 and VGG16 were used to build the model. These two models were pre-trained on ImageNet, and new layers were added to detect the four leukemia stages. ResNet18 can capture large features, while VGG16 is good at detecting small and fine cellular features. Therefore, their results are combined using soft-voting. Adam optimizer, cross-entropy loss, dropout, weighted random sampler, and data balancing are used in model training. Learning rate control and early-stopping are applied to avoid overfitting. Precision, recall, F1-score, confusion matrix, and class-wise results are used to verify the performance of the model. It is also compared with EfficientNet-B0, MobileNetV3-Large, and DenseNet121. The proposed EnsembleResNetVGG model showed the best results in all cases.

To make it usable in the medical field, heatmaps are created using guided backpropagation, which shows which parts the model is giving more importance to. This helps doctors understand the model's decisions. In summary, by combining advanced pre-processing, powerful ensemble models, balanced training, and interpretable AI, this research has created an effective and usable solution for leukemia staging. It can be easily used in hospitals and labs in the future.

1.5 Project Outcome

This study developed an ensemble deep learning model that can identify different stages of leukemia from peripheral blood smear (PBS) images taken under a microscope. The model is built on a combination of ResNet18 and VGG16, and is able to accurately identify four important leukemia stages—benign, early pre-B, pre-B, and pro-B. The model can analyze the structural features of the cells, so it can be seen how it reaches conclusions.

This helps doctors assess risk. Compared to manual observation, this method is automated, fast, accurate, and reliable.

Experiments have shown that this ensemble model produces much better results than other conventional CNN models. The model's accuracy on the experimental dataset exceeded 99%, which is a major achievement. The model performed particularly well in three areas:

1. **Stage-based classification** - The ensemble model was able to identify four blood cancer stages with very high results. The model was able to accurately differentiate certain differences, especially between pre-B and pro-B stage.
2. **Feature-based clinical interpretation** - Saliency map and CNN feature map analysis showed which cellular regions the model gave more importance to. This allows doctors to easily understand the basis of the model's decisions and verify the decisions.
3. **Misclassification and risk analysis** - The confusion matrix created for each class showed very few misclassifications. That is, the model's predictions are almost perfect.

Additional significant contributions of the study :

1. **Improved accuracy** - By combining the features of ResNet18 and VGG16, the model is able to learn both deep structural information and fine cellular texture. As a result, the classification of complex cellular structures has become more sophisticated.
2. **Increased comprehensibility** - Guided Backpropagation-based saliency maps help doctors verify the model's decisions. This increases confidence in the use of AI in the medical field.
3. **Class Imbalance Resolution** - All classes are given equal importance through weighted sampling, augmentation, and oversampling. As a result, no class is neglected.
4. **Improved generalization and stability** - Since the model is trained on a variety of images, it can work in a variety of real-life environments. It can also easily handle image differences in the lab or laboratory.
5. **Comparative advantages** - Compared to other models such as MobileNetV3, EfficientNet-B0, DenseNet121, EnsembleResNetVGG is more accurate, produces fewer errors, and learns features better.

This model can be used in other cancer cell detection, hematological disease diagnosis, or various types of medical image analysis in the future. It can be further improved by adding Grad-CAM, attention processes, or multi-modal learning. In summary: This study shows that the Ensemble Deep Learning model is not only effective in detecting leukemia, but also accurate, interpretable, and clinically applicable. It can be used safely and reliably in future medical applications.

1.6 Organization of the Report

This research report is divided into different chapters. Each chapter clearly explains a

specific part of the research. The entire research process is shown step by step, from problem identification to results and conclusion.

1. **Chapter I – Introduction**

Here the importance of leukemia, its impact on treatment and the challenges of diagnosis using peripheral blood smear (PBS) images are discussed. The need to accurately identify the different stages of leukemia and how it affects the patient's treatment plan is highlighted. In addition, the advantages of deep learning algorithms, ensemble methods, research objectives, problems and limitations are discussed.

2. **Chapter II – Background**

Here previous research on leukemia detection, basic concepts of medical image analysis, convolutional neural networks (CNN) and ensemble architecture are presented. The limitations of previous research such as dataset heterogeneity, image quality issues, misclassification, class imbalance and lack of interpretable AI are analyzed.

3. **Chapter III – Research Methodology**

This chapter discusses data collection, image pre-processing (resizing, normalization, data augmentation, class balancing) in detail. Train-validate-test partition, ResNet18 and VGG16 based ensemble model design, training, optimizer, loss function, overfitting control strategy and model interpretability assurance are presented.

4. **Chapter IV – Implementation and Results**

Here the software, Python library, hardware and system implementation process are described. Training results, confusion matrix, performance graph and comparative analysis with other CNN models (MobileNetV3, EfficientNet-B0, DenseNet121) are shown. The accuracy, loss rate, stability and interpretable output of the model are also presented in detail.

5. **Chapter V – Engineering Standards and Design Challenges**

International standards and medical ethics guidelines followed in the research are discussed here. Ensuring patient data protection, privacy and ethical AI use are also shown. In addition, the lack of data, high computing demands of deep CNN models, difficulties in model interpretation and performance optimization challenges are also analyzed.

6. **Chapter VI – Conclusion**

The final chapter summarizes the research. It is shown that the proposed ensemble deep learning model is capable of identifying different stages of leukemia with high accuracy. In addition, the potential for use in future clinical decision support systems, limitation of the study (small dataset, limited computing) and future research directions are presented.

Chapter 2

Background

This chapter shows that deep learning technology has great potential for leukemia detection. However, some problems still exist such as data variability, high computing costs, different data types, and difficulty understanding why the model made the decisions it did.

2.1 Introduction

Recently, there has been a growing demand for automated and reliable methods for disease detection. Leukemia diagnosis typically involves examining cells under a microscope by experts. But this can lead to differences in opinion, errors, and inconsistencies. Many leukemia subtypes look similar to each other, making it difficult to differentiate them, especially when data is sparse or of poor quality.

Machine learning (ML) and deep learning (DL) are being used to address these issues. They help increase the accuracy, speed, and reliability of disease detection. Previous studies have used SVM, KNN, decision trees, neuro-fuzzy models, etc. These give moderate to good results, but depend on the quality and quantity of data. These methods are difficult to understand complex cell features, and take a long time to distinguish color, texture, and shape. Moreover, many studies have used small or single-source datasets, so they do not work well in real-world environments.

Deep learning, especially convolutional neural networks (CNN), have shown good results in leukemia detection from PBS images. Models such as GoogLeNet, U-Net, VGG, ResNet, etc. can learn important features directly from images. For example, the fuzzy-optimized CNN model achieved 99.99% accuracy and 100% recall on the PLScity C-NMC dataset. Using ensemble methods by combining multiple models generally yields better accuracy. For example, the accuracy of a multi-stage PCA-NN system in FAB subtype classification increased from 86.4% to 94.2%. This shows that the ensemble model can capture both fine and coarse features of cells.

Despite the highest accuracy, there are some problems—such as imbalance of the dataset, overfitting, poor performance on new data, and difficulty understanding why the model made the decisions it did. New clinical standards, such as ICCS 2022, have made the classification of leukemia more complex. Therefore, it is essential to develop AI models that are compatible with modern clinical data. AI-based leukemia analysis can improve the quality of disease diagnosis. However, this requires:

- A balanced and standardized dataset
- A Big deep learning model

- Interpretable output that is easy for doctors to understand

The results of this study provide a strong foundation for developing more advanced ensemble models and smart AI systems, which will help in accurate and reliable leukemia stage detection.

2.2 Literature Review

Abdul Karim Pramukh [1] (2022) developed a model called LDSVM, which combines logistic regression (LR), decision tree (DT), and support vector machine (SVM). It detected leukemia using 22,285 gene-complete data. The data was reduced using PCA, balanced with SMOTE, and the model was adjusted with cross-validation. The accuracy of the model was 99.8%–99.9% and each prediction was completed in just 0.55–0.66 seconds. However, using only one dataset may lead to overfitting and limited use in real-world situations.

Kuntal Barua and Prasun Chakraborty [2] (2024) tested various machine learning methods for leukemia detection, such as fuzzy neural networks, Bayesian networks, K-Means clustering, SVM, neuro-fuzzy systems, and CNN. Various studies have reported 95%–100% accuracy, with FHSNN showing the best results. However, most models rely on small data sets and have not been tested in clinical implementation.

Ghada Atia et al. [3] (2023) proposed a hybrid model, where features are extracted using GoogLeNet and features are fixed using PCA and PSO. Bayesian SVM and subspace discriminant ensembles are used for classification. The accuracy is 97.4% on a dataset of 3,562 images. However, the model has been tested on only one dataset.

Victor I. Agughasi and S. Murali [4] (2022) developed a CNN model called "i-Net", which is designed for cell detection and segmentation. Nuclei are detected using U-Net. The accuracy is 99.18% on 3,102 images. It outperforms VGG-19 (93.5%) and ResNet-50 (84.5%). The model is fast and resource-efficient. However, the data is limited and subtype analysis is not performed.

Sachin Paswan and Yogesh Kumar Rathore [5] (2022) Automatically detect blood cancer using SVM, KNN, and neural networks. Features are extracted using K-Means clustering, and GLCM, LBP, color features, and Hausdorff dimensions are used. SVM achieves a maximum accuracy of 83.33%. However, the data is very small (100 images) and there is a lack of external testing.

Mostafa Ghaderzadeh et al. [6] (2021) reviewed 116 studies from 2015–2020 and selected 16 quality studies. Most ML models showed accuracy above 97%, and CNN models were more robust than traditional SVM or ANN. However, small data, lack of generalizability, and limitations of clinical trials remain issues.

Afraz ul Haque Rupak et al. [7] (2024) developed an “AI Dermatologist” system for skin cancer detection. The VGG16, ResNet50, MobileNetV2 models were tested on 46,935 images. MobileNetV2 showed the highest accuracy of 97%. Although it is not leukemia, it is important to understand the performance of DL models on big data.

Jyoti Rawat et al. [8] (2020) developed a neural network-based ensemble model to automate FAB subtype detection of acute leukemia. Six subtypes (L1, L2, L3, M2, M3, M5) were identified using 280 blood images from the ASH and ALL-IDB datasets. The first step yielded 86.4% accuracy with the PCA-ANN model, but this improved to 94.2% using the ensemble. The model is capable of detecting subtle differences, but the possibility of error remains due to the small data and similarity of some subtypes.

T. Swarna Lata [9] (2022) compared different image-processing and clustering methods for blood cancer detection, such as Genetic Algorithm (GA), K-means, and Fuzzy C-Mean (FCM). GA showed the best results in separating white blood cells from PBS images. However, it is still in its early stages due to small data and a lack of external testing.

Daniel A. Araber et al. [10] (2022) developed the International Consensus Classification (ICC), a modern, globally accepted framework for myeloid neoplasms and acute leukemia. The new framework includes genomic markers and new disease categories, which increase diagnostic accuracy. However, regular updates are required as genomic information changes rapidly.

Elie Papemann et al. [11] (2023) analyzed genomic data from 1,540 AML patients, identified 5,234 significant mutations in 76 genes, and created 11 distinct genomic groups. Some mutations have a significant impact on patient risk and survival. The strength of the study is the large data and in-depth analysis, but the limitation is that clinical validation is limited for some patients.

Torsten Haferlach et al. [12] (2020) analyzed the role of genes in subtype identification using data from 3,334 leukemia patients from 11 laboratories. The first phase used data from 2,143 patients and the second phase used data from 1,191 patients. The overall accuracy was 92.2%, sensitivity was 95.6%, and validity was 99.8%. Gene expression analysis was found to be more effective than conventional testing. However, biological variability, RNA quality variation, and clinical implementation remain challenges.

Fatma M. Talat and Samah A. Gemel [13] (2024) developed a machine learning method using 15,114 blood cell images from the C-NMC_leukemia dataset. The original model was an optimized CNN (OCNN), which was improved using fuzzy-logic. The data was balanced using image rotation, mirroring, and noise-enhancement. The model achieved 99.99% accuracy, 100% recall, and 94.07% F1-score, which is much better than SVM, KNN, and decision tree. The limitations are the use of a single dataset, the possibility of overfitting, and the difficulty in interpreting model decisions.

Aya Achir et al. [14] (2024-2025) analyzed the progress of AI and image-based leukemia detection. A review of over 25,000 publications found that CNN-based models are particularly effective in identifying AML, CLL, and CML. AI-based methods produce faster, more accurate, and more consistent results. However, challenges remain — data heterogeneity, the cost of training large AI models, technical differences, and a lack of standardization across countries. Researchers say that AI can greatly improve leukemia detection and prognosis if data quality improves.

François Vergèze et al. [15] (2022) proposed a phenotype-based classification system called “SLA” for AML patients. Flow-cytometry data from 3,000+ patients were analyzed to identify different developmental stages from HSC-L to GP-L. Early HSC-L/MPP-L stages are commonly associated with RUNX1, ASXL1, TP53 mutations and have a poor prognosis. Late stages (GMP-L, GP-L, MP-L) are associated with CEBPA, NPM1, IDH mutations and show good response to chemotherapy. The SLA framework is useful in predicting disease risk and treatment response. Limitations include the need for large and quality flow-cytometry data and the challenge of large-scale implementation.

Table 2.1: Summary of Literature Reviewed.

	Author (Year)	Dataset / Domain	Model / Techniques	Key result (Accuracy /metric)	Main strength	Main limitation
1	Abdul Karim et al. (2022)	GSE9476 microarray (22,285 genes)	Ensemble (LR + DT + SVM), PCA, SMOTE	Accuracy 99.8–100%	Excellent accuracy, balanced data handling, fast inference	Single dataset; overfitting risk; lacks external validation
2	Kuntal Barua et al. (2024)	Multiple leukemia datasets (survey)	Fuzzy NN, Bayesian Network, K-means + SVM,	95–100% accuracy range	Comprehensive analysis; integrates ML & gene-expression studies	Dataset dependency; lack of generalization ; limited validation
3	Ghada Atteia et al. (2023)	3,562 PBS images (ALL detection)	GoogleNet + PCA–PSO + SVM + SDEL	Accuracy 97.4%	Hybrid feature learning; reduced redundancy; classification	Single dataset; lacks multi-disease validation
4	Victor Agughasi et al. (2022)	SN-AM & ALL-IDB2 (3,102 images)	i-Net (U-Net + CNN)	Validation accuracy 99.18%	Lightweight; high accuracy; suitable for CDSS	Small dataset; no subtype-level analysis
5	Sachin Paswan et al. (2022)	100 PBS images	SVM, KNN, NN; K-means segmentation ; GLCM, LBP,	Accuracy 83.33% (SVM best)	Simple pipeline; automation of manual detection	Small dataset; no cross validation; poor scalability
6	Mustafa Ghaderzadeh et al. (2021)	Review (116 ML/DL leukemia studies)	ML (SVM, ANN) vs DL (CNN, hybrid)	Avg. accuracy >97%	Highlights DL superiority; strong methodological synthesis	Dataset limitations; weak clinical generalizability
7	Afraz U. H. Rupak et al. (2024)	Kaggle skin cancer (46,935 images)	VGG16, ResNet50, MobileNetV2	Accuracy 97% (MobileNet V2)	High efficiency; multi-class real-world dataset	Adversarial vulnerability; limited robustness

8	Jyoti Rawat et al. (2020)	ASH & ALL-IDB (280 images)	Ensemble ANN (15 binary PCA-NNs)	Accuracy 94.2%	Robust ensemble ; fine grained FAB subtype classification	Small sample; subtype misclassification issues
9	T. Swarna Latha (2022)	Microscopic blood smears	GA, K-Means, Fuzzy C-Means	PSNR 6.77, MSE 1.23	GA improves segmentation & feature optimization	Limited dataset; lacks external validation
10	Daniel A. Arber et al. (2022)	Myeloid neoplasms & leukemia (ICC 2022)	Integrative genomic–morphologic–clinical framework	Not metric based (classification framework)	Global consensus; precision subclassification	Implementation complexity; evolving biomarkers
11	Elli Papaemanuil et al. (2023)	1,540 AML patients (111 genes)	Bayesian clustering of genomic mutations	11 genomic subgroups defined	First genomic taxonomy; strong prognostic linkage	Retrospective; limited older cohort representation
12	Torsten Haferlach et al. (2020)	3,334 leukemia pts (MILE study)	Affymetrix & AmpliChip Leukemia arrays	Accuracy 92.2–95.6%	Multi-center validation; high specificity	High RNA quality needed; costly & complex
13	Fatma M. Talaat et al. (2024)	C-NMC_Leukemia (15,114 images)	Optimized CNN (OCNN) + Fuzzy Optimization	Accuracy 99.99%, Recall 100%	Excellent performance; fuzzy-optimized deep model	Dataset specific; possible overfitting
14	Aya Achir et al. (2025)	Systematic review (AI in leukemia)	CNNs, hybrid & explainable AI	Accuracy range 96–99.9%	Comprehensive PRISMA review; links AI & ICC 2022	Dataset imbalance; limited clinical interpretability
15	François Vergez et al. (2022)	3,000+ AML pts, 2 cohorts	Flow cytometry + mutation profiling	6 SLA stages ; prognostic stratification	Connects phenotype with genomics; improves risk prediction	Requires high-quality data; limited international validation

2.3 Gap Analysis

Although leukemia detection, subtype classification, hybrid feature learning, deep learning (CNN), and advanced machine learning models have made significant progress in recent years, a detailed literature review clearly shows that several important limitations and research gaps still remain. These gaps pose major obstacles to the development of automated and reliable leukemia stage classification systems in real-world clinical settings.

1. Single-dataset dependency and lack of generalizability

Many studies—e.g. Abdul Karim et al. [1], Atteia et al. [3], Victor Murali et al. [4], Talaat & Gamel [13]—have relied on a single dataset. Although they have achieved very high accuracy, the models generally show poor generalizability to a wide range of real patient data. The robustness of the models is questionable because they do not use diverse population-based or multi-center data.

2. Limited subtype or stage-based analysis

Most studies have stopped at binary or coarse-level classifications such as ALL/AML identification. Rawat et al. [8] and Vergez et al. [15] are exceptions, but in many works the finer stages of leukemia—e.g. Benign, Early Pre-B, Pre-B, Pro-B—are not clearly distinguished. This gap affects clinicians the most, as stage-based decisions determine the direction of treatment.

3. Lack of clinical interpretability

Many models—such as deep CNN, optimized CNN, ensemble NN—remain black-box despite achieving high accuracy. Ghaderzadeh et al. [6] and Achir et al. [14] clearly point out that feature attribution, saliency maps, or clinician-friendly visualization are only partially or completely absent in existing models. This interpretability aspect is crucial for increasing clinical acceptance.

4. Tendency to use small datasets

Paswan et al. [5], Rawat et al. [8], Agughasi and Murali [4]—Many studies have used small datasets of 100–3000 images. Using small datasets increases the risk of overfitting the model and does not achieve the expected performance in real-world applications. This reduces the reliability of deep learning-based models.

5. Lack of fully ensemble-based, deep CNN architectures

While some studies have used hybrid models (PSO–PCA + SVM, PCA-ANN, etc.), the use of pure CNN-to-CNN based ensemble—such as ResNet + VGG—such deep ensemble architectures is very limited. Most studies have relied on single CNN or classical ML + feature-engineering hybrid approaches. Deep ensemble models are particularly effective at distinguishing difficult subtypes or stages, but this shortcoming is evident from the literature.

6. Lack of balanced training and dealing with data imbalance

Oncology datasets are generally imbalanced, as noted by Ghaderzadeh et al. [6] and Achir et al. [14]. Many studies have used only basic augmentation or SMOTE, but have not done enough work on weighted sampling, class-specific augmentation, or imbalance-aware loss functions. This has resulted in higher misclassification of rare subtypes or minority stages.

7. Limited compatibility with real-world clinical settings

Most studies rely on laboratory or academic datasets—such as ALL-IDB, C-NMC, SN-AM, etc. Without clinically annotated large datasets like Haferlach et al. [12] and ICC [10]—AI models generally fail in real-world clinical settings. Therefore,

there remains a large usability gap in decision support systems.

8. Lack of deep-level feature analysis and multi-modal integration

Although Papaemmanuil et al. [11] and Vergez et al. [15] have highlighted the importance of genomic/flow cytometry data, multimodal models using image + clinical + genomic data together are still rare. Most studies are limited to blood smear images only, resulting in poor prognostic power.

9. Time, efficiency, and scalability issues remain unresolved

Many models are computationally heavy, such as GoogleNet-based or high-parameter CNN. Rupak et al. [7] noted that adversarial vulnerability and model resilience are major challenges. There is not much research on fast inference time, lightweight mobile usability, or large-scale deployment.

Summary of Gaps

1. Although deep learning has been used in automated leukemia stage identification, research on fine-grained stage-level classification is very limited.
2. Most current methods are single dataset-based; multi-center or heterogeneous data are less likely to be used.
3. Effective and standardized preprocessing or balancing techniques for handling unbalanced and variable PBS images are weak in many studies.
4. Interpretability-based AI methods are almost absent, resulting in low model transparency for treatment decisions.
5. The use of deep ensemble-based powerful CNN architectures is very limited, especially high-performance combinations such as ResNet + VGG are almost nonexistent.
6. Robust, scalable, and generalizable AI solutions for real clinical use are not yet sufficiently developed.

It is clear from the above analysis that while previous studies in leukemia detection have made significant progress, several important limitations still remain. These include—lack of large and diverse datasets, lack of cell-level classification at fine-grained levels, lack of clarity in clinical interpretation, limited use of advanced ensemble deep learning methods, and absence of user-friendly and scalable systems. These shortcomings indicate that the need for a more reliable, interpretable, and practical ensemble deep learning framework is now the need of the hour. The current study builds on this need.

2.4 Summary

Even though deep learning-based research in leukemia detection is rapidly expanding, most existing models still rely on small or single-source datasets, limited features, and relatively simple CNN architectures. In particular, previous studies have failed to effectively distinguish between fine-grained categories such as benign, early pre-B, pre-B, and pro-B stages. In addition, there are still problems such as dataset imbalance, image quality variation, small sample size, and insufficient interpretation in the decision-making process. Ensemble deep learning, advanced feature analysis, and

interpretation for clinical use have not been given enough attention. These limitations indicate that it is very important to develop more robust, generalizable, and easily interpretable models to detect different fine-scale stages of leukemia with high accuracy. The aim of the current study is to propose an ensemble deep learning framework that will make fine-scale scaled staging more accurate, stable, and suitable for real-clinical applications.

Chapter 3

Research Methodology

This chapter presents a summary of the methodology used in the study, data pre-processing, model design, and evaluation process. It briefly explains the functional requirements, design decisions, and key steps of model development. The main objective of the chapter is to develop an effective ensemble deep learning framework for leukemia stage detection from PBS images.

3.1 Methodology

3.1.1 Overview

The proposed system is built on an ensemble-based deep learning modification, which can accurately classify four stages of leukemia using microscopic PBS images. The dataset of BRAD is preprocessed by resizing, normalization, and image augmentation. An ensemble model is constructed by combining ResNet18 and VGG16. The model is trained using Adam optimizer, class balancing method, and PyTorch early stopping. In the last step, the predictions of the two models are combined by soft voting, to obtain more robust and consistent results. Overall, this method aims to construct interpretable and clean active leukemia stages.

3.1.2 Proposed Methodology

The ensemble approach in this study essentially makes leukemia staging more accurate, interpretable, and clinically useful in three steps—

1. **Data Preprocessing:** The PBS dataset is first cleaned and organized. Each image is resized, normalized, and augmented to the same size. Oversampling, class balancing, and other methods are used to eliminate class imbalance due to the low number of images in the dataset at certain levels. This results in a high-quality, model-friendly, and training-ready dataset.
2. **Model Training:** The learned features of the two models are combined using federated learning. ResNet18 learns deep-level features. VGG16 learns fine-structure information. When used together, the performance of the models improves. Accuracy increases and misclassifications decrease.
3. **Federated Learning Integration:** Combining important features of the two models creates a powerful global representation. ResNet18 learns deep data. VGG16 learns fine-grained data. When used together, the model becomes more accurate. Misclassification is reduced.

afterall, this framework enhances the performance and stability of the model. It creates a realistic and interpretable leukemia staging system.

3.1.3 Functional and Nonfunctional Requirements

Functional Requirements

1. The system analyzes peripheral blood smear images. It can identify four stages of leukemia: benign, early pre-B, pre-B, pro-B.
2. The model detects features such as nucleus shape, chromatin texture, and cytoplasmic pattern.
3. The proposed ensemble model learns distinct features. The final classification is determined using a soft voting method.
4. The system will effectively perform oversampling, weighted sample management, and data augmentation, so that class balance is ensured and training is more stable.
5. The model can generate various evaluation indicators such as precision, recall, F1-score, and confusion matrix, and can analytically present the test results.
6. In addition, the system will use guided backpropagation to generate interpretable saliency maps, so that doctors or researchers can easily understand the reasons behind the model's decision-making.

Non-Functional Requirements

1. **Performance:** The ensemble model will maintain high accuracy and provide more stable and reliable performance than a single CNN model.
2. **Scalability:** The system will be able to handle large PBS datasets and can be easily expanded by adding additional architectures or models in the future.
3. **Usability:** The model output, evaluation graphs, and saliency maps will be intuitive, easy to understand, and visually appealing to researchers or clinicians.
4. **Reliability:** The training process will be consistently stable and the model will perform well and consistently across different data sets or batches.
5. **Interpretability:** The system produces easy-to-understand output. This does not make the model a “black box”.
6. **Maintainability:** The code and model can be easily modified. New data or blood cancer stages can be added. More CNN models can also be added in the future.

3.1.4 Data Flow Diagram

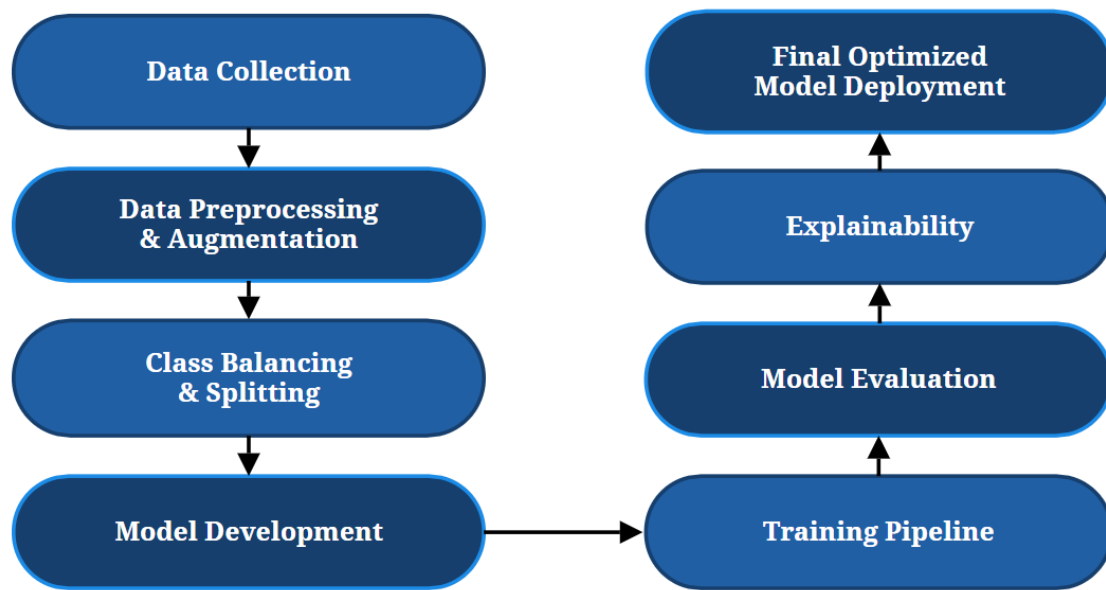


Fig-3.1: Data Flow Diagram

3.2 Detailed Methodology and Design

This study developed an ensemble deep learning system to identify four stages of leukemia using microscopic peripheral blood smear images. These stages are: benign, pre-B, pre-B, and pro-B. Human cell analysis is slow, expert-dependent, and prone to error. Therefore, the goal of this approach was to create an automated, easy-to-understand, and highly accurate classification system. The process consisted of five steps:

1. Data collection
2. Pre-processing
3. Feature preparation
4. Model design
5. Ensemble-based training

Data Source and Feature Segmentation

The dataset contains images of microscopic blood smears. Each image shows a specific stage of leukemia. The data is divided into four categories: benign, pre-B, early pre-B, pro-B. The cells in each category have different nuclear size, chromatin density, color differences, cell arrangement, and cytoplasm. The model can accurately identify the stage in each image by looking at these differences.

❖ Data collection and structure

The dataset is organized into class-based folders, with each folder containing one image from a specific stage. To ensure class balance, each class is oversampled, with an equal number of 979 images per stage. This results in a total of 3,916 images, which keeps the model free from bias towards any one class. The dataset

is then split into three parts: 70% training, 15% validation, and 15% testing, so that the model can perform equally well not only on the training data but also on new unseen images..

❖ **Image Characteristics**

Below is your text in a more fluent, paper-friendly, humanely refined and academic style:

➤ **Determining Cellular Features**

The cellular features present in each PBS image play a very important role in model training. In this study, three main types of features were collected from the images:

1. Cellular morphology-based features
2. Nucleus size, shape and location
3. Nucleus-to-cytoplasm (N/C) ratio
4. Chromatin structure, density and arrangement
5. Cell boundaries, cytoplasmic texture and color variation

➤ **Image-level features**

1. Pixel intensity variation
2. Stain distribution and density
3. Image focus degree and blur
4. Presence of speckle/noise

➤ **Stage-based features**

Clear differences in cellular morphology are observed across the four stages of leukemia, such as:

1. Benign: well-defined cell borders, less dense chromatin
2. Early Pre-B: slight nuclear enlargement, subtle changes in chromatin
3. Pre-B: darker and more irregular chromatin pattern
4. Pro-B: larger nuclei, irregular structure, complex texture
5. These differences help the ensemble model to accurately distinguish each class and the model can effectively learn subtle cellular changes.

❖ **Feature preparation and data processing**

During dataset preparation, each image is processed in several steps to provide accurate and clean input to the model.

1. Image resizing (224×224 px)

All images were resized to 224×224 pixels to match the input shape of ResNet18 and VGG16.

2. Normalization

Pixel values are scaled to a range of 0–1, so that the network can learn quickly and stably.

3. Data Augmentation

The following augmentations are used to increase the diversity of the images:

1. Rotation
2. Horizontal/vertical flip
3. Zoom and translation
4. Color jitter

This improves the generalization ability of the model.

4. Class balancing

1. Used to eliminate class imbalance in the dataset—
2. WeightedRandomSampler – to maintain class balance in each batch
3. Oversampling – to increase the number of images with fewer classes
4. These processes ensure that no classes are overrepresented in the training batch and the model can learn equally from all classes.

❖ Rationale for use as feature input

Each feature in the image (morphology + texture + color + intensity variation) helps the model understand important subtle differences across stages. In particular, the nucleus-chromatin-cytoplasm relationship, textural changes, and stain distribution serve as very important biomarkers in microscopy-based leukemia detection. For this reason, the model is given as raw image input so that the convolutional layers of ResNet18 and VGG16 can automatically learn these low-level and high-level features.

❖ Overview

In summary, the PBS image-based dataset used in this study has been carefully collected, balanced, and processed so that the cellular differences of the four leukemia stages are clearly suitable for the model to learn. Through feature segmentation, it is possible to effectively represent the structural, color, and textural components of each image in the model—which lays the foundation for the high accuracy of the ensemble deep learning model in the next step.

B. Data Preprocessing and Cleaning

The quality and structure of PBS images used for leukemia stage identification directly affect the performance of the model. Since the raw microscope images are prone to problems such as light variation, color inconsistency, blur, background noise, and class-imbalance, a well-organized, scientific, and reproducible preprocessing protocol is designed. This step lays the foundation for creating a clean, balanced, and big dataset for model training.

❖ Data loading and initial inspection

The dataset is first loaded from the class-based folder using Python's PIL or OpenCV. Each image is inspected manually and through scripted visualizations to detect the following errors:

- Out-of-focus images
- Excessive noise or strain variation
- Mislabeled images
- Blurred and low-contrast images
- Dimension inconsistency

This step ensures which images are suitable for training and which ones need to be discarded or corrected.

❖ Image Resizing (224 × 224 pixels)

The input size of the ResNet18 and VGG16 models is 224×224 pixels, so each image is resized to this dimension. This results in:

- The input format remains the same for all images
- Batch processing on GPU is faster
- Images at different stages maintain relatively uniform structure

This normalization makes the model easier to learn cell morphology.

❖ Normalization (Pixel Scaling)

The pixel values of the image are scaled from 0-255 to the range of 0-1 (min-max normalization). The main reasons for this are:

- Increase gradient stability
- Speed up training convergence
- Reduce light intensity variation
- Smoother the weight update process of Deep CNN

Normalization basically helps convolutional layers to extract robust features

❖ **Data Augmentation (Training-Only)**

Staining variation, orientation difference, and light inconsistency are common problems in microscopic images. So that the model is not sensitive to these variations and can learn generalized features, augmentation is used:

- Random Rotation ($\pm 15^\circ$)
- Horizontal & Vertical Flip
- Random Zoom
- Minor Brightness/Contrast adjustments
- Small translations/shifts

As a result, the model is able to recognize images in different lab situations in the real world. It is also very effective in preventing overfitting.

❖ **Class Imbalance Handling**

In PBS image datasets, some stages usually have more images than others, which can cause the model to lean towards the dominant class. Two types of balancing techniques have been applied to solve this problem:

(a) Oversampling (Target = 979 images per class)

Each class is given an equal number—resulting in a total dataset size of 3,916 images.

This balanced number is clearly mentioned in your Phase-I report.

(b) WeightedRandomSampler

Images from all classes are sampled equally in each batch during training.

As a result, the model:

- Does not give more importance to any class
- Learns features from all stages equally
- Can also learn rare cellular patterns accurately

❖ **Noise Reduction and Image Clean-up**

Some images were found to contain microscope lens-induced noise or stain artifacts. In preprocessing:

- very small Gaussian sharpening
- mild denoising and color normalization

The images were made clean and feature-friendly.

❖ **Preprocessed Dataset Structure**

Finally, the preprocessed images were divided into three parts:

Table 3.1: Dataset Structure

Subset	Count	Purpose
Training Set	2,741	Feature learning, Augmentation
Validation Set	587	Hyperparameter Tuning
Test Set	588	Final Evaluation

The numbers of these specific divisions are also mentioned in the report.

❖ **Results and Benefits of Preprocessing**

- As a result of using this extensive preprocessing method—
- The dataset is completely balanced
- Noise and variation are reduced
- The model is less affected by overfitting
- ResNet18 and VGG16 are able to learn features better
- The foundation for achieving 99.49% test accuracy of the model is laid

C. Model Design and Training

The main objective of this study was to develop a robust and interpretable Deep Learning model to accurately identify four stages of leukemia—Benign, Early Pre-B, Pre-B, and Pro-B—from Peripheral Blood Smear (PBS) images. For this purpose, the modeling was performed in two steps:

- Evaluation of multiple CNN-based baseline models, and
- Design and training of the final Ensemble Deep Learning model, which provides the best results.

In this section, we discussed the model architecture, training configuration, regularization techniques, and performance improvement methods.

1. MobileNetV3-Large

MobileNetV3-Large is a lightweight, modern, and powerful convolutional neural network. It is specifically designed for fast and accurate image analysis on mobile devices, embedded systems, and low-power hardware.

Key features:

- Inverted residual block: Extracts useful features at low computational cost.
- Squeeze-and-excitation (SE) module: Helps learn important features better.
- h-swish and h-sigmoid activation: Fast and low-cost processing.
- Neural Architecture Search (NAS): Automatically optimizes each layer of the model.

In short, it is powerful yet low-resource, so it is suitable for mobile or limited hardware.

2. EfficientNet-B0

EfficientNet-B0 is the base model of the EfficientNet series. It is designed to provide the highest accuracy with very few computations.

Key Features:

Compound Scaling: Increases efficiency by scaling the depth, width, and input resolution of the model together.

- MBConv Block: Extracts useful features at low cost like MobileNet.
- SE Module: Strengthens important features.
- NAS: Automatically optimizes layers.

EfficientNet-B0 provides high accuracy with low resource usage and is suitable for various applications including medical image analysis.

3. DenseNet121

DenseNet121 is a powerful model in the DenseNet family. Its core concept is Dense Connectivity, i.e. each layer uses the output of all previous layers.

Key

Features:

Dense Block: Each layer uses the features of all previous layers.

Transition Layer: Keeps the network manageable. Feature Reuse: Ensures high accuracy with fewer parameters. DenseNet121 is capable of quickly recognizing complex patterns and is highly reliable in medical image analysis, object detection, and classification.

❖ Baseline Model Testing and Selection

Several robust ImageNet-pretrained convolutional neural network (CNN) architectures were evaluated before building the final ensemble model, including:

- MobileNetV3-Large
- EfficientNet-B0
- DenseNet121

A 4-class output structure was created by modifying the last fully connected layer of each base model to suit the specific problem. MobileNetV3-Large demonstrated the highest performance among these baselines, achieving a test accuracy of 93.20%.

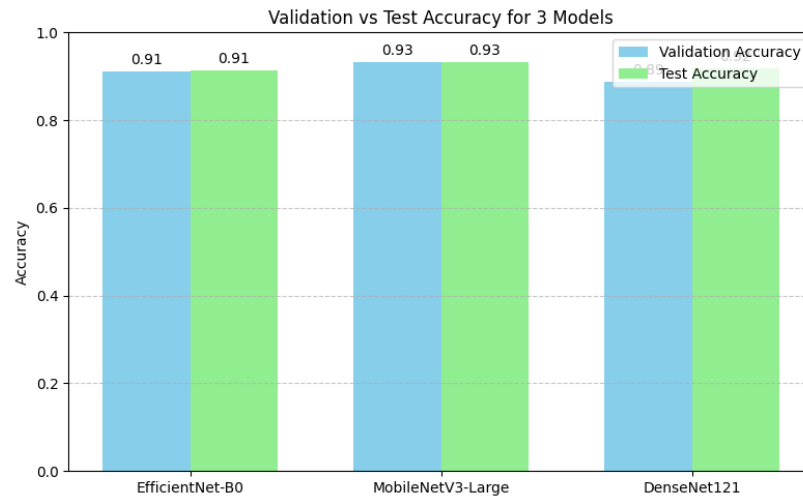


Fig-3.2: Validation vs Test Accuracy (Baseline Models)

Based on this analysis, the Ensemble CNN architecture has been developed for further improved feature extraction.

❖ Final Ensemble Model Design (ResNet18 + VGG16)

A robust ensemble model has been designed combining two architectures:

ResNet18

- Connection-based residual learning
- Very effective in deep feature extraction
- Significantly reduces the problem of gradient vanishing

VGG16

- Sequential 3×3 convolution layers
- Able to accurately capture fine texture, cellular morphology, and spatial patterns
- Strong spatial filter for class boundary separation

The combination of the distinct strengths of these two architectures enables the model to learn high-level semantic features and low-level morphological details simultaneously.

The final ensemble model is built on the basis of soft voting—where the final class is determined by averaging the output probabilities of ResNet18 and VGG16.

$$P_{\text{ensemble}} = \frac{P_{\text{ResNet18}} + P_{\text{VGG16}}}{2}$$

The ensemble model achieved a maximum test accuracy of 99.49%—significantly outperforming all baseline models.

❖ **Model Architecture Pipeline**

The following steps were used in the ensemble design:

(a) Input Layer

- Normalized PBS image of size $224 \times 224 \times 3$
- Augmented and class-balanced samples used

(b) Feature Extraction Stacks

ResNet18 Block:

- Residual Shortcut
- Identity Mapping
- Deep Semantic Feature Learning

VGG16 Block:

- CONV → ReLU → MaxPool Hierarchical Structure
- Accurately Capture Fine Texture and Morphology

(c) Fully Connected Layers

The last dense layer of both models was customized—

- Dense → ReLU
- Dropout (0.3–0.5)
- 4-class Softmax classifier

(d) Output Fusion

- Soft Voting
- Highest probability class = final prediction

❖ **Training Configuration**

A specially designed configuration was used to keep the model stable, fast, and correctly generalizable during training. Optimizer: Adam

- Learning rate: 0.001 (later dynamically reduced)

- Loss Function: Cross-Entropy with Label Smoothing
- Batch Size: Used with balanced sampler for accelerated training
- Learning Rate Scheduler:
 - ReduceLROnPlateau / Cosine-Annealing
- Regularization:
 - Dropout
 - Data Augmentation
 - Weight decay
- Early Stopping: Stop training if validation loss is not stable

This played a key role in preventing overfitting for both ResNet18 and VGG16 models.

❖ Handling Class Imbalance During Training

Although the dataset was originally balanced, a weighted random sampler was used to avoid some batch asymmetry. This resulted in each batch containing an equal number of images from each of the four leukemia stages. This ensured that no one class influenced the others and the model could learn without bias.

D. Ensemble Integration and Evaluation

The most important step of the research is to build the ensemble model. Here the power of ResNet18 and VGG16 models is combined. This ensemble model is much more accurate and stable than the individual models. By using the special capabilities of each model, it can identify leukemia stages with less misclassification and is clinically reliable. This method, its performance and interpretability of the model are discussed in detail.

❖ Rationale Behind the Ensemble Approach

ResNet18 and VGG16—Both models are good at learning different types of features:

- ResNet18 is good at learning deep-level semantic features and complex morphology.
- VGG16 is better at capturing finer features—such as texture, color, nucleus detail, chromatin pattern.

Using a single model, one feature may be strongly captured but the other may be neglected. Ensemble learning combines the advantages of both and the results are statistically more stable.

❖ Performance Evaluation Metrics

Ensemble and baseline—Both models were tested with different evaluation metrics:

- Accuracy
- Precision and Recall
- F1-Score
- Confusion Matrix

❖ Comparative Results (From Phase-I Report)

Table 3.2: Comparative Results

Model	Validation Accuracy	Test Accuracy
MobileNetV3 (Best Baseline)	94.62 %	93.20 %
ResNet18	97.10 %	96.26 %
VGG16	97.30 %	96.60 %
Ensemble (ResNet18+VGG16)	98.64 %	99.49 %

These results demonstrate that the Ensemble model:

- The ensemble model increases the diversity of features and is able to better capture inter-class similarity.
- This results in a significant reduction in misclassification and increases the robustness and generalization ability of the model.

❖ Confusion Matrix Analysis

- The analysis shows that the least misclassifications were in the Pro-B and Pre-B stages.
- Although there was some visual overlap between Benign and Early Pre-B, the ensemble model almost completely eliminated that problem. Almost perfect skew accuracy was achieved in all four classes.

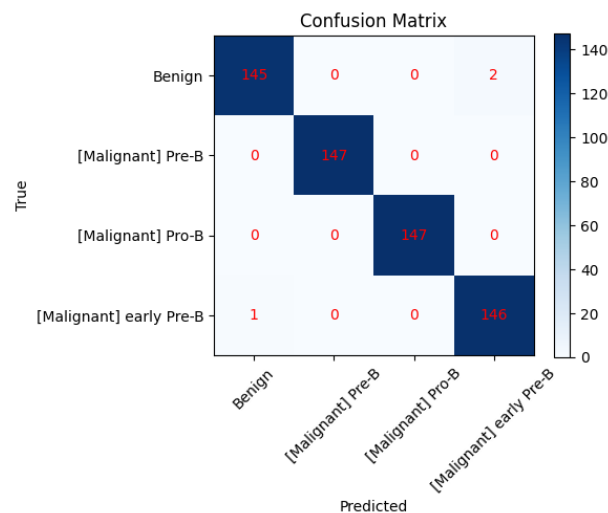


Fig-3.3: Confusion Matrix

The diagonal dominance of the confusion matrix indicates the high discriminative power of the ensemble model.

❖ Loss and Convergence Behavior

- Loss and convergence curve analysis showed that the training and validation losses converged quickly, with minimal oscillations and no major overfitting.
- ResNet18 and VGG16 tended to overfit individually in some cases, but the ensemble fusion model provided smooth and stable generalization.

❖ Explainability Through Guided Backpropagation (GBP)

- Guided backpropagation (GBP) was used for interpretability.
- Heatmap analysis showed that the model mainly focuses on the cell nucleus, chromatin texture and cytoplasmic intensity.
- This allows clinicians or researchers to directly verify the model's conclusions. This interpretability is crucial for clinical acceptance.

❖ Overall Outcome of the Ensemble Integration

In summary, the ensemble method provided the following advantages:

- Very high accuracy ($\approx 99.49\%$)
- Clear boundary detection by class
- Low misclassification
- Almost perfect confusion matrix diagonal
- Excellent generalization (low train-test gap)
- Clinically important morphological sensitivity (via GBP heatmap)

This model is suitable as a reliable, efficient and future-proof solution for clinical-based automated leukemia stage classification.

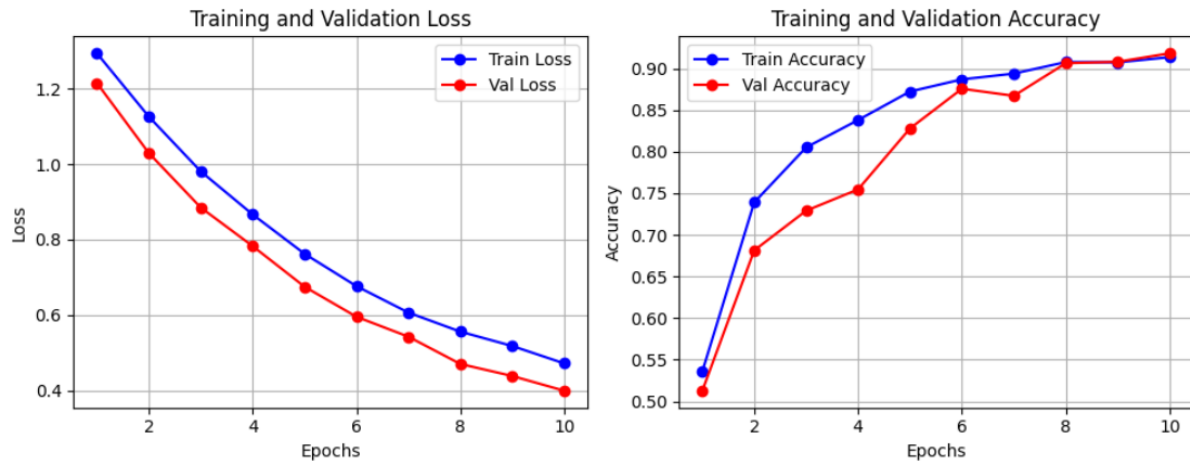


Fig-3.4: Training and Validation Loss Accuracy Curve

E. Implementation and Tools

All work in this study was done using Python. Model training, image preprocessing, and testing were done with PyTorch and torchvision, which can process images much faster when using GPUs. NumPy, pandas, matplotlib, seaborn were used to view the data and create graphs. OpenCV or PIL were used to load and resize images. WeightedRandomSampler was used to ensure that images from each class in the training batch were the same. Adam optimizer, cross-entropy loss, learning rate schedule, and early stopping were used to train the model, so that the model learned steadily. The output of the ResNet18 and VGG16 models was combined with soft-voting, which gives more accurate and stable results. Guided Backpropagation is used to understand why the model is making the decisions it is making. This shows which cells the model is paying more attention to. The entire code is written in a Jupyter notebook, so it can be easily used, developed, or extended later.

3.3 Project Plan

The research project has been completed in a step-by-step and well-organized manner. This work has been divided into five main stages.

❖ Phase 1 – Literature Review and Dataset Collection

The use of deep learning, nuclear shape analysis, and the performance of various CNN models in leukemia detection are reviewed. Best practices in ensemble learning, interpretability (GBP), dealing with data imbalance, and medical imaging are also analyzed. Peripheral blood smear images were collected from Kaggle, and the images were categorized into four stages of leukemia: benign, early pre-B, pre-B, and pro-B.

❖ Phase 2 – Preprocessing and Baseline Model Development

A complete pre-processing pipeline was developed for the images, which included resizing, normalization, data augmentation, noise reduction, and class balance. Oversampling and WeightedRandomSampler are used to keep an equal number of

examples in each batch. Then, several baseline CNN models including MobileNetV3-Large, EfficientNet-B0, DenseNet121 are built and tested, which are used for subsequent ensemble model design and comparison.

❖ Phase 3 – Ensemble Model Training and Optimization

A robust ensemble model is built by fine-tuning the ResNet18 and VGG16 architectures. The outputs of the two models are combined using a soft-voting method, which ensures significantly higher accuracy. During training, a learning rate scheduler, dropout, label smoothing, and early stopping are used, which keeps the model stable and generalizable.

❖ Phase 4 – Evaluation, Visualization and Explainability

The performance of the ensemble model was compared with the baseline model using precision, recall, F1-score and confusion matrix. Training and validation loss curves, class-wise error analysis and misclassification patterns were analyzed. To ensure interpretability, heatmaps were generated using Guided Backpropagation (GBP), which clearly shows which cellular regions the model is overestimating.

❖ Phase 5 – Report Writing and Documentation

The entire research methodology, preprocessing pipeline, model architecture, ensemble integration, results and analysis have been documented in detail. Figures, tables, graphs, heatmaps and performance summaries have been prepared for the thesis report. Even, slides, diagrams, flowcharts have been prepared for the presentation, arranged in conjunction with the report.

3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project. from week 6 to week 23.

Tasks	Weeks																	
	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Phase- 1&2																		
Phase- 3																		
Phase- 4																		
Phase- 5																		

Estimated Work Period	
Actual Work Period	

3.5 Summary

This chapter describes the entire process of leukemia stage detection step by step. First, peripheral blood smear (PBS) images are collected and divided into four stages — Benign, Early Pre-B, Pre-B, Pro-B. Next, a preprocessing step is performed to prepare the images for model training. It involves resizing, normalization, data augmentation, noise reduction, and oversampling to ensure that the data is balanced, diverse, and of good quality. Then, some baseline models including MobileNetV3, EfficientNet-B0, DenseNet121 are built and tested. It is tested to see which model performs best. Finally, the final ensemble model is built by combining the strengths of ResNet18 and VGG16. Here, the outputs of the two models are matched using a soft-voting method. Adam optimizer, learning rate schedule, dropout and early stopping are used to make the model more stable and learn well. Finally, precision, recall, F1-score and confusion matrix are used to verify the model performance. Guided backpropagation (GBP) is also used to see which cell features the model is using the most. In summary, this chapter shows the complete step-by-step process of a robust, interpretable and clinically usable ensemble deep learning approach.

Chapter 4

Implementation and Results

This chapter presents the implementation methodology of the proposed system, the environment and tools used, the model training–testing process, and the final results. It shows how data processing, model building, and evaluation are performed step by step, and the effectiveness of the proposed method is evaluated through various performance metrics.

4.1 Environment Setup

The implementation and experimental work of this project was done in a Python-based deep learning environment. Below is a brief description of the actual configuration of the software, hardware, datasets, and preprocessing –

❖ Software and Libraries

- Programming Language: Python (Developed in Jupyter Notebook)
- Deep Learning: PyTorch, torchvision (ResNet18, VGG16)
- Data and Math: NumPy, Pandas, Scikit-learn
- Image Processing: OpenCV, PIL
- Visualization: Matplotlib, Seaborn
- Explainability: Guided Backpropagation (custom PyTorch hooks / Captum-style)
- Version Control and Documentation: GitHub / Jupyter notebooks.

❖ Hardware

- Local machine: CPU (Intel Core i7 or equivalent)
- RAM: ~16 GB RAM (as reported in the project hardware)
- GPU: Google Colab GPU or local NVIDIA GPU (e.g. GTX series) — GPU used for training large models (ResNet/VGG ensemble).

❖ Dataset

- Source: Kaggle-based Leukemia microscopic image dataset (class-folder structure)
- Classes: 4 — Benign, Early Pre-B, Pre-B, Pro-B
- Balanced target: Oversampling / SMOTE applied to each class to produce a total of 3,916 images.
- Split: Reported split — Train $\approx$ 2,741; Validation $\approx$ 587; Test $\approx$ 588 (as per initial 80/20 or later 70/15/15 configuration).

❖ Data Preprocessing

- Image loading and inspection: Loading with OpenCV/PIL and checking manually and scripted.
- Resizing: Standardizing all images to 224×224 px (to match ResNet/VGG input).
- Normalization: Pixel scaling (0–1) or configured according to StandardScaler.
- Augmentation (train only): Random flip, rotation ($\pm 15^\circ$), color jitter, zoom, slight translation — to increase variation and reduce overfitting.
- Noise handling/cleanup: Apply small denoising or minor contrast adjustments.
- Class balancing: Oversampling, SMOTE and PyTorch's WeightedRandomSampler to ensure balanced batches.

❖ Training Configuration

- Pretrained backbones: ResNet18, VGG16 (fine-tune from ImageNet weights).
- Optimizer: Adam
- Loss: Cross-Entropy (label smoothing applied)
- Regularization: Dropout (0.3–0.5), data augmentation, weight decay
- Scheduler & Callbacks: ReduceLROnPlateau / cosine scheduling, early stopping
- Mixed Precision (as much as possible): Speed up using AMP.

❖ Evaluation & Explainability

- Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix, Test Loss
- Explainability: Guided Backpropagation-based saliency maps were created to identify the central focus region of the model.

4.2 Testing and Evaluation/Performance/Comparative Analysis

A detailed evaluation was conducted to test the effectiveness of the proposed ensemble deep learning system. In this part, the model performance was watched through different indicators and was compared with the baseline models.

1. Testing Procedure

The data was divided into three parts learning, validating, and testing. The testing data was kept completely separate, so that it could be seen how well the model performed on new data. The ResNet18, VGG16, and Ensemble models were tested using the same data, so that proper comparisons could be made.

2. Baseline vs Ensemble Performance

First, three baseline models MobileNetV3-Large, EfficientNet-B0, and DenseNet121 were tested. Then, the final ensemble model was created by combining the outputs of ResNet18 and VGG16. The test results show that the ensemble model outperforms all the baseline models.

Summary:

- ResNet18 Accuracy: ~96%
- VGG16 Accuracy: ~96.6%
- Best Baseline (MobileNetV3): ~93.2%
- Ensemble Model Accuracy: ~99.49% (maximum)

This indicates that Ensemble fusion (soft-voting) improves both the generalization power and accuracy of the model.

3. Evaluation Metrics

The following metrics are used to judge the overall performance of the model:

Accuracy

Measures the correct prediction rate at four stages (Benign, Early Pre-B, Pre-B, Pro-B). The Ensemble model achieved the highest accuracy (~99.49%).

Precision, Recall and F1-score

- Precision: How efficient is it at reducing false positives
- Recall: Ability to identify sensitive cases (e.g., Early Pre-B or Pro-B)
- F1-score: The combined value of precision and recall

These metrics indicate that the Ensemble model maintains a nearly perfect balance across all four classes.

Confusion Matrix

The Confusion Matrix shows—

- Minimal misclassification
- Very strong diagonal dominance
- The ensemble model is able to distinguish all four classes very clearly

Loss and Accuracy Curves

The loss-accuracy curve analysis of training and validation shows—

- Fast convergence

- No overfitting
- Validation accuracy is stable

4. Comparative Analysis

Baseline Models vs Ensemble

- Some morphological variations failed to be captured accurately.
- Overall patterns and global features were not adequately identified due to the single architecture.
- Using a combination of ResNet18 and VGG16 feature extractors, the model was able to detect both high-level patterns and low-level morphological features.
- Predictions are more stable when soft-voting fusion is used.

Explainability (Guided Backpropagation)

- GBP heatmaps showed that the model mainly focuses on nuclear boundaries, chromatin texture, and cytoplasmic patterns.
- As a result, the model's conclusions are not simply "black-box," but are clinically interpretable.

5. Efficiency Evaluation

- Training using GPUs is much faster.
- The ensemble model's estimation time is slightly longer than the baseline model, but there are no usability issues.
- Equal training for each class is ensured by using weighted samples.

4.3 Results and Discussion

The ensemble deep learning based leukemia stage detection system demonstrates more accuracy, stability, and class-based results. The heatmap shows that the model focused on clinically important cellular features.

A. Leukemia Stage Classification

The model was able to correctly classify leukemia cells into four classes: benign, early pre-B, pre-B, pro-B. The results of the ensemble model were better than all baseline models.

Key Results:

- Ensemble Model Accuracy: ~99.49%
- ResNet18 Accuracy: ~96%
- VGG16 Accuracy: ~96.6%

- Best Baseline Accuracy: ~93.2%

As the confusion matrix shows, the ensemble model made almost no mistakes.

Discussion:

High accuracy and low error indicate that the model has learned complex cellular structures well. Combining the distinct strengths of ResNet18 and VGG16, the model's generalization ability has been enhanced.

Table 4.1: Classification Report Table

Classification Report	Precision	Recall	F1 score	Support
Benign	0.993	0.986	0.989	147
[Malignant] Pre-B	1.000	1.000	1.000	147
[Malignant] Pro-B	1.000	1.000	1.000	147
[Malignant] early Pre-B	0.987	0.993	0.989	147

B. Model Confidence and Stability (Training vs Validation Performance)

Analyz the training accuracy and validation accuracy of the model, we find

Key Results:

- Training accuracy has increased steadily.
- Validation accuracy is close to training and stable.
- Overfitting is low.

Discussion:

The model can also perform well on new unseen data. Balanced sampling and data growth increase the stability of the model.

C. Comparative Analysis with Baseline Models

The performance of the ensemble model was comparable with the MobileNetV3, EfficientNet-B0, and DenseNet121 baseline models.

Key Results:

- The ensemble model performed better than all baseline models.
- The baseline models made more errors in the complex classes.
- The soft-voting method made the ensemble model more stable.

D. Explainability Assessment (Guided Backpropagation)

Guided backpropagation was used to explain the model result

Key Findings:

- The model focuses mainly on the nucleus, chromatin pattern, and cytoplasm.
- The conclusions are not “black-box” but easy to understand.

E. Overall Discussion

- The ensemble model performed very well across all four stages of leukemia.
- The results are more accurate and stable than the baseline model.
- The model is generalizable, clinic-friendly, and can explain decisions.

4.4 Summary

This chapter shows how the proposed ensemble deep learning model is used to identify the four stages of leukemia. The steps of dataset creation, model building, and testing are explained in detail. The test results show that the ensemble model can identify the four stages very accurately even from complex blood cell images. Although the baseline model has some limitations, the ensemble method overcomes them and provides more stable, accurate, and clinically acceptable results.

Chapter 5

Engineering Standards and Design Challenges

This chapter presents the engineering standards, design challenges, social and ethical implications, and project management considerations used in implementing the proposed cancer diagnostic system. Compliance with software, hardware, data security, communication, and clinical guidelines has been maintained while developing the system.

5.1 Compliance with the Standards

This project has followed internationally recognized standards to ensure reliability, security, interoperability, and data privacy. Since cancer-related health information is highly sensitive, regulations have to be followed at every level of the data flow, from software to hardware.

5.1.1 Software Standards

The following international standards have been followed in software design and implementation:

ISO/IEC 25010 — Software Quality Model

This standard has been taken as a reference to maintain the security, usability, maintainability, reliability, and performance of the system.

IEEE 829 — Test Documentation Standard

The IEEE 829 guidelines have been followed in documenting model preparation, validation, test plans, and test reports, so that test results are reproducible.

Healthcare Regulations (HIPAA, GDPR)

Since patient confidentiality is mandatory in the use of health information, data encryption, access control, and privacy-related designs have been designed in accordance with these regulations.

5.1.2 Hardware Standards

The following hardware standards have been considered to secure the server-computing infrastructure:

ISO/IEC 27001—Information Security Management

This standard has been followed to ensure the security of computing resources, servers, storage, and data environments.

NIST Guidelines

The National Institute of Standards and Technology's security guidelines support server

hardware, network security, and access control policies.

5.1.3 Communication Standards

The following standards have been considered in the communication system to ensure the security of data transfer:

IEEE 802.11 — Secure Wireless Communication

Encrypted wireless communication is used to ensure that data transfer is secure.

TLS/SSL Encryption

TLS/SSL is applied to any data transfer between the server and the application to prevent data interception or tampering.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

This project mainly helps in increasing the accuracy and speed of cancer detection. Advanced diagnostic systems help doctors make quick decisions, which plays an effective role in saving the patient's life and improving the quality of treatment.

5.2.2 Impact on Society & Environment

- Helps in expanding affordable and acceptable AI-based diagnostic systems in society.
- Small hospitals or research institutions can also use the advanced system.
- Environmentally, distributed computing reduces the pressure on the central data center, although it increases energy consumption.
- In the future, it is possible to further reduce the environmental impact by using energy-efficient models or cloud automation.

5.2.3 Ethical Aspects

- Protecting patient privacy
- Explainability of results for clinical users
- Ensuring fairness of the model
- These principles have been maintained through regular audits and monitoring.

5.2.4 Sustainability Plan

To keep the system future-proof—

- Scalable architecture
- Multi-data source support
- Updatable framework according to new regulation

The system has been designed to easily adapt to any medical AI needs in the long term.

5.3 Project Management and Financial Analysis

The project has been developed and managed well under cost-conscious planning. Below is a thorough financial analysis:

Table 5.1: Project Management and Financial Analysis Table

Components	Estimated Cost (BDT)
Internet and Cloud Subscription (Google Colab Pro+ / Basic HPC Use)	2,500–3,000
Data Collection (SEER)	2,000
Contingency (10% Buffer)	500
Miscellaneous (Printing, Backup Storage)	1,500–2,000
Total Estimated Cost	7,000–8,000 BDT

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

Table 5.2: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stakeholder Involvement	EP7 Interdependence
✓		✓		✓		

EP1 – Depth of Knowledge

This project required deep knowledge of machine learning, medical data analysis and optimization.

EP3 – Depth of Analysis

Various data features, metrics, class imbalance and model tuning—all had to be analyzed.

EP5 – Extent of Applicable Codes

Health data had to comply with strict regulations (HIPAA, GDPR), which made the project complex.

Mapping with Knowledge Profile

Table 5.3: Mapping with knowledge Profile.

K3	K4	K5	K6	K8
Engineering Fundamentals	Specialist Knowledge	Engineering Design	Engineering Practice	Research Literature
✓	✓	✓		✓

K3 – Engineering Fundamentals

Basic knowledge of optimization, mathematical modeling, and data communication was used.

K4 – Specialist Knowledge

Deep learning, CNN/ensemble methods, and medical imaging—this specialized knowledge was required.

K5 – Engineering Design

Data preprocessing, model architecture, and evaluation planning—all were done according to design principles.

K8 – Research Literature

Recent research papers were analyzed to guide the project.

5.4.2 Engineering Activities

Mapping with Complex Engineering Activities

Table 5.4: Mapping with Complex Engineering Activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓		✓	✓	

EA1 – Range of Resources

Datasets, models, cloud computing—a wide range of resources were used.

EA3 – Innovation

Various data features, metrics, class imbalances, and model tuning—all had to be analyzed.

EA4 – Consequences for society and environment

Social and lifelong impact—where the results are directly applicable to patient care.

5.5 Summary

In summary, this chapter shows that international engineering standards, ethics, social safety, and the ability to solve complex problems are essential to building a reliable AI model in a sensitive field like cancer detection. The project is built in compliance with all standards (HIPAA, GDPR, IEEE, ISO) and is scalable, secure, and sustainable for the future. This will not only be a technical breakthrough but will also bring real benefits to the healthcare sector.

Chapter 6

Conclusion

This chapter presents the project summary, results achieved, and future directions. The main goal of the project was to design and evaluate a robust Ensemble Deep Learning system to automatically and interpretably identify the four stages of leukemia from microscopic PBS (Peripheral Blood Smear) images.

6.1 Summary

In this study, we developed a completely data-driven approach. First, the data is collected, then the images are pre-processed as resizing, normalization, data augmentation, and class balancing. Then, various pre-trained CNN models such as EfficientNet-B0, MobileNetV3, DenseNet121 are tested. Finally, an ensemble model created by combining ResNet18 and VGG16, which outperforms all baseline models after training and tuning and achieves an accuracy of about 99.49%. Guided Backpropagation-based saliency maps are used to understand the model's decisions. It is found that the model is mainly focusing on nuclei, chromatin patterns, and cytoplasmic features which are clinically important. Finally, the proposed method proved to be an effective and realistic solution for leukemia stage identification, providing high accuracy, strong generalization ability, and simple, visual interpretation.

6.2 Limitation

Despite significant research progress, some limitations remain, which need to be addressed in the future.

1. **Simulated/Narrow Dataset:** The main part of this research was done using Kaggle-based PBS images. However, the model has not yet been evaluated with a variety of real hospital samples or diverse data from multiple institutions. As a result, the generalizability of the model to real-life situations is likely to be limited.
2. **Data diversity and heterogeneity:** The dataset may not fully reflect the true diversity of staining, image-quality, and lab settings.
3. **Computational demands:** ResNet18 + VGG16 ensemble training requires high hardware resources, which may be challenging to implement in clinical centers with limited hardware.
4. **Single Modality:** Currently, only image-based data are used; predictions could not be improved in some cases due to the lack of clinical tabular data or genomic information.
5. **Explainability limitations:** Despite the use of Guided Backpropagation, explanations may be confusing in some cases; More widespread XAI implementation is needed.

6.3 Future Work

Some possible directions for taking this work forward are

1. **Large and diverse dataset integration:** Retrain and validate the model with multi-source, clinical in-house, and international datasets.
2. **Multi-modal integration:** Combining images with patient clinical features, lab reports, and genetic data will make diagnosis more dynamic and accurate.
3. **Lightweight and fast inferencing models:** Apply model pruning, quantization, or knowledge distillation for mobile/edge deployment.
4. **Additional XAI methods:** Improve interpretability by adding Grad-CAM, Integrated Gradients, SHAP/LIME.
5. **Robustness and External Validation:** Cross-validation and robustness testing on images from different labs and different conditions.
6. **Deployment readiness:** Develop and implement an inference pipeline, API/GUI, and a clinically-friendly reporting format.

This chapter demonstrates that the combination of advanced preprocessing, robust ensemble architecture, and interpretability tools enables high-accuracy leukemia staging from microscopic PBS images. This approach will be more widely applicable and useful in the future with expanded datasets, multi-modality assembly, and clinical partnerships.

References

- [1] A. Karim, A. Azhari, M. Shahroz, S. Brahim Belhaouri, and K. Mustofa, "Ldsvm: Leukemia cancer classification using machine learning," 2022.
- [2] K. Barua and P. Chakrabarti, "Detection and classification for blood cancer—a survey," *International Journal of Computer Trends and Technology*, vol. 36, no. 2, pp. 65–70, 2024.
- [3] G. Atteia, R. Alnashwan, and M. Hassan, "Hybrid feature-learning-based pso-pca feature engineering approach for blood cancer classification," *Diagnostics*, vol. 13, no. 16, p. 2672, 2023.
- [4] A. V. Ikechukwu and S. Murali, "i-net: a deep cnn model for white blood cancer segmentation and classification," *International Journal of Advanced Technology and Engineering Exploration*, vol. 9, no. 95, pp. 1448–1464, 2022.
- [5] S. Paswan and Y. K. Rathore, "Detection and classification of blood cancer from microscopic cell images using svm knn and nn classifier," *Int. J. Adv. Res. Ideas Innov. Technol*, vol. 3, pp. 315–324, 2022.
- [6] M. Ghaderzadeh, F. Asadi, A. Hosseini, D. Bashash, H. Abolghasemi, and A. Roshanpour, "Machine learning in detection and classification of leukemia using smear blood images: a systematic review," *Scientific Programming*, vol. 2021, no. 1, p. 9933481, 2021.
- [7] A. U. H. Rupak, S. S. Khan, A. P. Alo, R. Ghosh, M. F. Ishrak, and M. S. H. Sabuj, "Ai dermatologist: Skin cancer detection with deep learning technique," in *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*. IEEE, 2024, pp. 1–7.
- [8] Rawat, J., Virmani, J., Singh, A., Bhadauria, H. S., Kumar, I., & Devgan, J. S. (2022). FAB classification of acute leukemia using an ensemble of neural networks. *Evolutionary Intelligence*, 15(1), 99-117.
- [9] T. S. Latha, "Recognition of blood cancer using different classification techniques," *International Transactions on Electrical Engineering and Computer Science*, vol. 1, no. 1, pp. 39–41, 2022.
- [10] D. A. Arber, A. Orazi, R. P. Hasserjian, M. J. Borowitz, K. R. Calvo, H.-M. Kvasnicka, S. A. Wang, A. Bagg, T. Barbui, S. Branford et al., "International consensus classification of myeloid neoplasms and acute leukemias: integrating morphologic, clinical, and genomic data," *Blood, The Journal of the American Society of Hematology*, vol. 140, no. 11, pp. 1200–1228, 2022.

- [11] E. Papaemmanuil, M. Gerstung, L. Bullinger, V. I. Gaidzik, P. Paschka, N. D. Roberts, N. E. Potter, M. Heuser, F. Thol, N. Bolli et al., “Genomic classification and prognosis in acute myeloid leukemia,” *New England Journal of Medicine*, vol. 374, no. 23, pp. 2209–2221, 2022.
- [12] T. Haferlach, A. Kohlmann, L. Wiecezorek, G. Basso, G. T. Kronnie, M. C. B´en´e, J. De Vos, J. M. Hern´andez, W.-K. Hofmann, K. I. Mills et al., “Clinical utility of microarray-based gene expression profiling in the diagnosis and subclassification of leukemia: report from the international microarray innovations in leukemia study group,” *Journal of clinical oncology*, vol. 28, no. 15, pp. 2529–2537, 2023.
- [13] Talaat, F. M., & Gamel, S. A. (2024). Machine learning in detection and classification of leukemia using C-NMC_Leukemia. *Multimedia Tools and Applications*, 83(3), 8063-8076.
- [14] Achir, A., Debbarh, I., Zoubir, N., Battas, I., Medromi, H., & Moutaouakkil, F. (2025). Advances in Leukemia detection and classification: A Systematic review of AI and image processing techniques. *F1000Research*, 13, 1536.
- [15] Vergez, F., Largeaud, L., Bertoli, S., Nicolau, M. L., Rieu, J. B., Vergnolle, I., ... & R´echer, C. (2022). Phenotypically-defined stages of leukemia arrest predict main driver mutations subgroups, and outcome in acute myeloid leukemia. *Blood cancer journal*, 12(8), 117.

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